Why we chose Multithreading: When we got the idea for the project, we predicted that, due to the upgrades in our game, we would need to check collisions between a large number of beams and asteroids, and that could potentially slow down our game by a lot, so we needed a way to implement this while making sure the game stays playable. We discovered that multithreading is how programs do several things in parallel, which was exactly what we needed.

What we learned: A thread is a separate part of the code which runs in parallel alongside the rest of it. While multiple threads can use the same data, changes may not be visible to all threads. We learnt that volatile and synchronized can make these changes visible to all the threads. Moreover, synchronized and locks prevent other threads from using the same block of code at the same time.

How we applied it: We use 3 threads in our code. One is the Swing thread, which we had to use by default and which is used to draw the screen. Another handles the rocket movement and the plasma beams, while the third handles the asteroid movement and all the collisions.

What we learned: While making the game, a bug occurred where the beams would not despawn after hitting an asteroid. We were initially using linked lists for all the objects and their hitboxes. The problem was that while one loop in a thread was iterating through the lists to check the collisions, the other was modifying the lists. A fix would have been a lock added everywhere the lists were used, but that meant that a lot of changes should have been made. Instead, we found that there exist collections built for concurrency, which can be used safely when the collection is used across multiple threads. We changed the hitboxes' linked list to a concurrent linked deque, which allowed beams to be removed while another thread was modifying the collection. We chose a concurrent linked deque because it allowed us to not make any other changes to the code.

Why we chose Github: When we got the idea for the project, we knew we would be sometimes be working and he same time, sometimes at different times and usually at different locations. When working from different locations and on different devices it's useful to use github because the changed someone made makes it clearer for the other person working on the same project, it also allows you to work on the most recent version without having to constantly send the other person the folder. It also allows people to work simultaneously on different things and then merge what they worked on and see were there are conflicts.

What we learned through research: We learned how to make branches which allows us to work on different things while also ensuring that the previous version still exists. We learned how to commit the changes we made to the main branch and how committing works. We also learned about merging branches.

How we applied it: We made a repository in github and then shared it so that we both have access to it (github repository link at the end of the file). We each worked in our own branch and committed changes when we were done working on it and included a message of what was changed.

What we learned: At the beginning we had not implemented github yet so we sent a zipped folder to the other person when we were done working on our coding, this cause some issues and made it disorganized, when we implemented github we learned just how much more organized it is instead of sending files to each other. We also learned how to connect Github to our coding app so that we could switch branches and commit changes to the branch through the coding app. A challenge we ran into at one point is that only the person who made the repository is able to make branches, while we did not find out how to allow both people to make branches, we got around this problem by deciding that each person has their own branch with their name. Github had a big impact in the end on our project as it ensured that we could see the other's commits, it kept things more organized, and it also ensured that we did not work on the same thing (which did happen before we switched to github).

Github link: https://github.com/LenaBruin/SpaceInvaders.git